

Preface to the Classics Edition

We are pleased that SIAM is republishing this book on the principles of computerized tomography. We enjoyed writing it and have heard from many people who found the book valuable. We're glad that the book is now back in print.

The worlds of tomography and medical imaging have not stood still in the twelve years since this book was first published, yet the basic algorithms presented in this book are just as important today as when they were first applied to tomographic imaging. The reconstruction formalism that underlies these algorithms helps us create efficient implementations for medical scanners and provides us with a theoretical framework to think about issues such as noise, sampling, resolution and artifacts. We hope that the new readers of our book will also appreciate the beauty of these algorithms.

The primary strength of this book consists of derivations that are appropriate for engineers and scientists who want to understand the principles of computerized tomographic imaging. We hope that these derivations will provide readers with insights to create their own implementations. For our readers more interested in the medical applications of computerized tomography we suggest they also consult Lee's [Lee98] or Haaga's [Haa94] books. For readers interested in more mathematical rigor or a discussion of the mathematical foundations of cross-sectional imaging we suggest they supplement our book with the book by Natterer and Wübbeling [Nat01].

The years since our book was first published have witnessed an increased interest in helical acquisition and reconstruction methods. We cover the basics of this topic in Section 3.6. Interested readers should also consult the paper by Crawford and King [Cra90] and the special issue of *Transactions on Medical Imaging* [Wan00] for an overview of current approaches.

Perhaps the most surprising development in the tomography field has been the application of these algorithms to geophysical phenomena. Researchers have used tomography to study the ocean [Mun95] at scales we are sure were not imagined by Hounsfield when he created his first scans. Likewise, tomographic algorithms have been used for whole-earth imaging [Iye93].

Lately, much work has used tomographic data (and MRI data) to generate volume renderings of internal organs. This work can allow physicians to “fly” through internal organs and provide interesting ways for radiologists to understand the state of a human body.

Before we wrote this book, it taxed our laboratory’s abilities to collect the data and create 64×64 reconstructions. Today, high-end medical scanners do helical cone-beam tomography, collect four rows of data (1000 detectors per row and 1000 views per rotation) in one-half second, and produce 512×512 images in near-real time. We hope our book will help people understand both ends of this spectrum.

The authors would like to thank the IEEE for their many years of support and for allowing us to republish our book. We would also like to thank our colleague Carl Crawford for his support over the years and his creative means of encouraging us to publish a new edition. One of the authors (MS) would like to publicly thank Barb Booth, Jim Ziegler, Gordon Speer, and Anne (Atkins) Burch for their early encouragement in the areas of mathematics, science, and publishing.

A number of corrections and some sample codes are available on the Web at <http://www.slaney.org/pct>.

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